

Name

Class



Year 9 Stage 5.1/5.2/5.3 Mathematics Assessment Task 4 2020

Instructions

- Reading time: 5 minutes
- Working time: 50 minutes
- Write in blue or black pen. Draw diagrams in pencil.
- Show all working where necessary. Marks may be deducted for poor or illegible writing.
- NESA Approved Calculators are permitted for use.
- A Reference Sheet has been provided with this examination.

Students in 9MATL, 9MATE and 9MATI *MUST* also attempt Section 3

Students in 9MATC, 9MATH, 9MATA, 9MATR and 9MATD *MAY* attempt Section 3.

Surface Area and Volume	Trigonometry	Probability	Total
/20	/20	/15	/55

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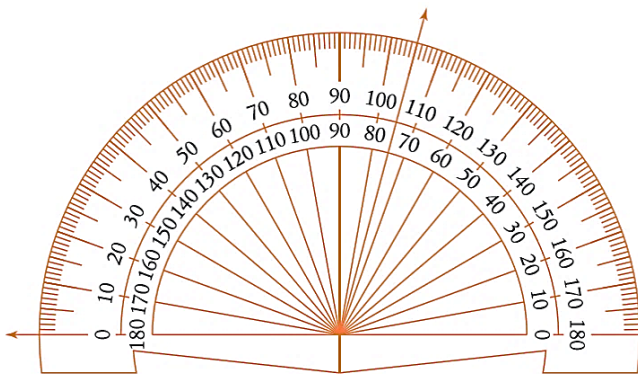
Multiple Choice (1 mark each) – CLEARLY circle the correct letter, A, B, C or D.

Surface Area and Volume

1. 64 g is equal to:

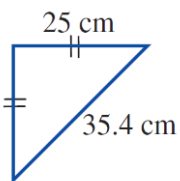
- A. 6.4 kg
- B. 0.64 kg
- C. 0.064 kg
- D. 64 000 kg

2. What is the limits of accuracy for the protractor below?



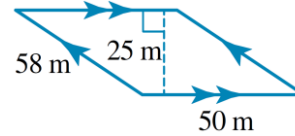
- A. $\pm 0.5^\circ$
- B. $\pm 1^\circ$
- C. 105°
- D. 75°

3. Find the perimeter of the shape below:



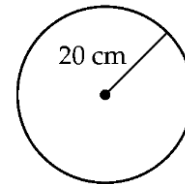
- A. 85.4 mm
- B. 95.8 mm
- C. 885 mm
- D. 442.5 mm

4. The area of the shape below is:



- A. 2900 m^2
- B. 1250 m^2
- C. 1450 m^2
- D. 72500 m^2

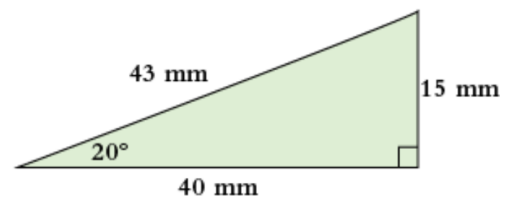
5. The area of the circle below is approximately:



- A. 125.66 cm^2
- B. 5026.55 cm^2
- C. 62.83 cm^2
- D. 1256.64 cm^2

Trigonometry

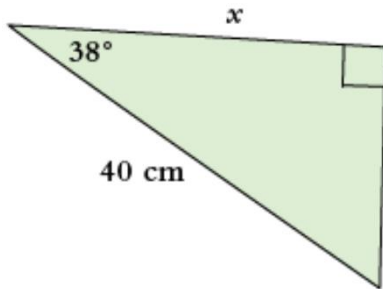
6. The correct labelling of the hypotenuse (H), opposite (O) and adjacent (A) sides in the triangle below is:



- A. H = 43 mm, O = 15 mm, A = 40 mm
- B. H = 40 mm, O = 43 mm, A = 15 mm
- C. H = 43 mm, O = 40 mm, A = 15 mm
- D. H = 40 mm, O = 15 mm, A = 43 mm

Section 1 continues on the following page

7. The equation that matches the information given in the triangle below is:

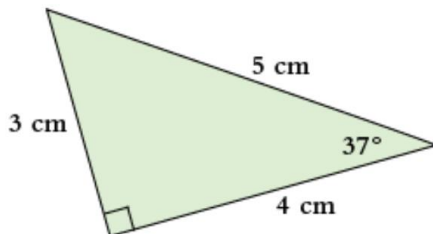


- A. $\cos 38^\circ = \frac{x}{40}$
- B. $\cos 38^\circ = \frac{40}{x}$
- C. $\sin 38^\circ = \frac{x}{40}$
- D. $\tan 38^\circ = \frac{40}{x}$

8. The value of $\tan^{-1}\left(\frac{2}{3}\right)$ to the nearest degree is:

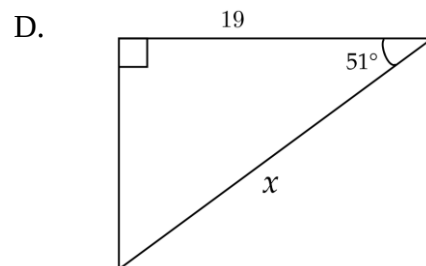
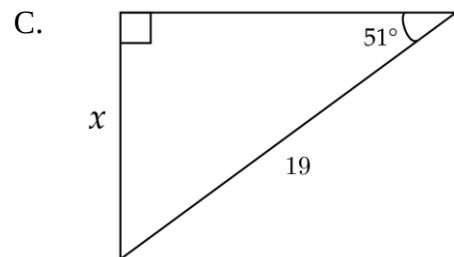
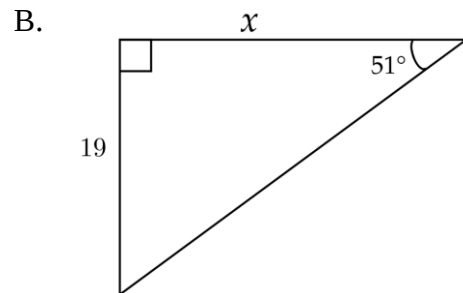
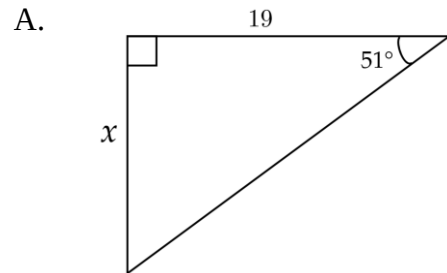
- A. 34°
- B. 42°
- C. 33°
- D. 0°

9. Find the value of $\sin 37^\circ$ as a fraction in the triangle below:



- A. $\frac{5}{3}$
- B. $\frac{4}{5}$
- C. $\frac{3}{5}$
- D. $\frac{3}{4}$

10. Given the equation $\tan 51^\circ = \frac{x}{19}$, which triangle is labelled correctly?



Section 1 continues on the following page

Probability

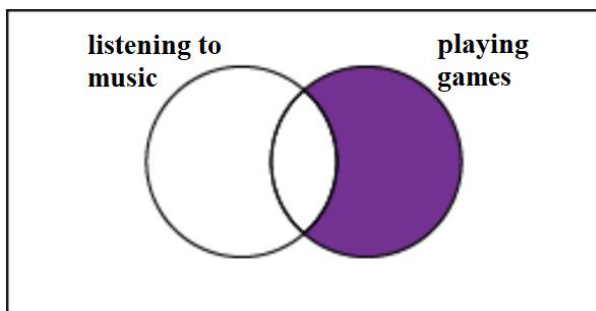
11. If a coin is flipped 300 times, the expected number of heads is:

- A. 300
- B. 600
- C. $\frac{1}{2}$
- D. 150

12. A six-sided die is rolled 150 times. A five was recorded 29 times. What is the experimental probability of rolling a five?

- A. $\frac{29}{150}$
- B. $\frac{121}{150}$
- C. $\frac{5}{29}$
- D. $\frac{1}{6}$

13. The Venn diagram below represents the results of a survey about the preferred relaxation activities of students. The shaded region represents:



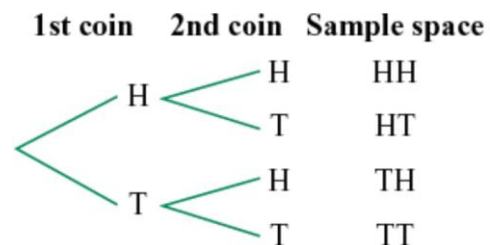
- A. people who like playing games but not listening to music
- B. people who like playing games and listening to music
- C. people who like listening to music
- D. people who like playing games

14. Students were surveyed about whether they played netball or soccer, results shown in the table below. What is the probability that a student chosen at random does *not* play either sport?

	Soccer	Not soccer	
Netball	10	6	16
Not netball	5	7	12
	15	13	28

- A. $\frac{11}{28}$
- B. $\frac{7}{28}$
- C. $\frac{7}{13}$
- D. $\frac{5}{12}$

15. Two coins are flipped. The sample space is represented in the tree diagram below. What is the probability of flipping *at least* one tail?



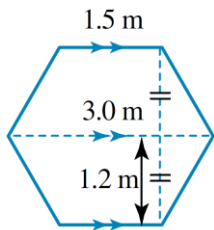
- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{3}{4}$
- D. $\frac{3}{8}$

End of Section 1

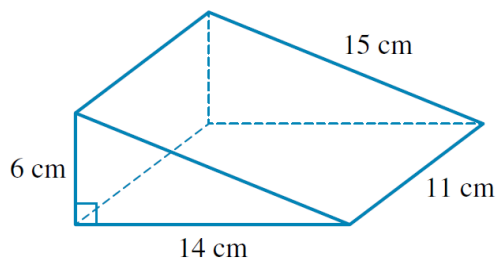
Short Answer Questions – Show all working.**Surface Area and Volume**

1. Calculate the area of the hexagon below.

2



2. For the triangular prism below, calculate:



(a) The surface area

2

(b) The volume

2

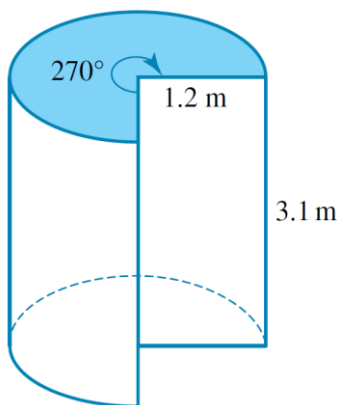
(c) Hence, find its capacity in litres.

1

Section 2 continues on the following page

3. Calculate the volume of the solid below.

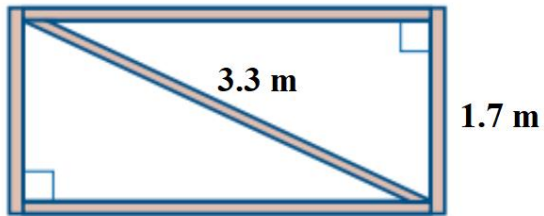
2



Section 2 continues on the following page

Short Answer Questions – Show all working.**Trigonometry**

4. A farm gate is 1.7 m high and is supported by a diagonal bar of length 3.3 m.



- (a) Find the width (horizontal) of the gate correct to 1 decimal place.

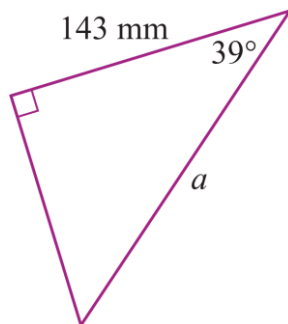
1

- (b) Find the angle between the diagonal bar and the horizontal correct to the nearest degree.

2

5. For the triangles below:

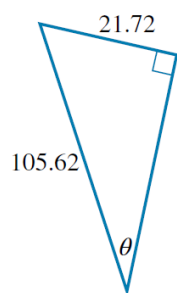
- (a) Find the value of a correct to 2 decimal places.

2

Section 2 continues on the following page

- (b) Find the value of θ to the nearest minute.

2



6. In $\triangle PQR$, $\angle Q$ is right-angled, $\angle R = 36.7^\circ$ and $QR = 5.2\text{m}$.

- (a) Draw and label a diagram representing this information.

1

- (b) Calculate the length of PQ , correct to 2 decimal places.

2

Section 2 continues on the following page

Short Answer Questions – Show all working.**Probability**

7. A bag contains a number of marbles of the colours blue, yellow, red and green. A student chooses a marble from the bag at random, records its colour and puts it back into the bag. She repeats this process 100 times. The results of the experiment are in the table below.

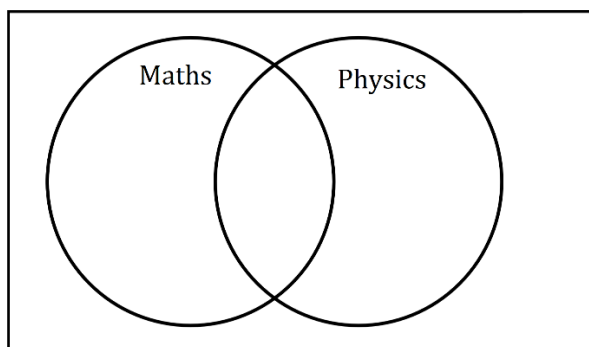
Colour	Blue	Yellow	Red	Green
Frequency	18	21	53	8

(a) What is probability that the next marble she chooses will *not* be green? 1

(b) If the bag contains 30 marbles, what is the expected number of red marbles in the bag? 1

8. In a group of 100 students, 63 study mathematics, 58 students study physics, and 30 students study both maths and physics.

(a) Represent this information in the Venn diagram below. 1



(b) One student is selected at random. What is the probability that the student studies neither mathematics nor physics? 1

9. A painter is selecting colours to paint the ceiling and walls of a room. He has three choices for the colour of the ceiling: Blue, White or Red. He has four choices of colour for the walls: Blue, White, Grey or Yellow.

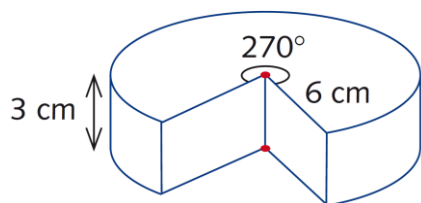
(a) Represent all the possible outcomes using an appropriate diagram or table. **1**

(b) If he chooses at random, what is the probability that the ceiling and walls are *different* colours? **1**

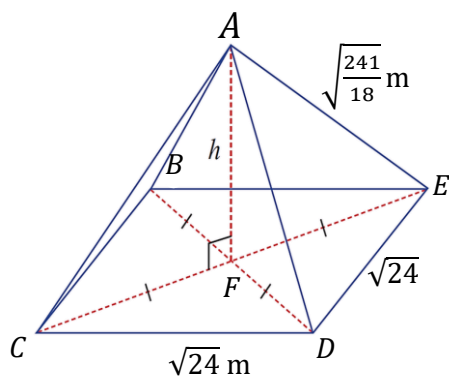
End of Section 2

Short Answer Questions – Show all working.**Surface Area and Volume**

1. A sector of cheese is cut from a cylindrical block of cheese. Find the surface area of the remaining piece, correct to two decimal places. 2



2. A pyramid has a square base BCDE of side length $\sqrt{24}$ m. The slanted edges of the pyramid are of length $\sqrt{\frac{241}{18}}$ m. Find the volume of the pyramid as a rationalised surd. 4

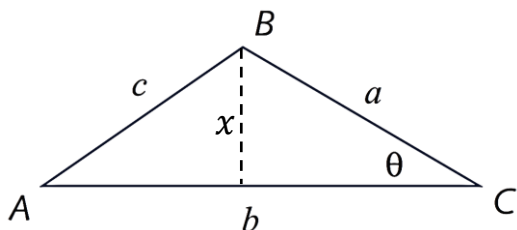


Section 3 continues on the following page

Short Answer Questions – Show all working.**Trigonometry**

3. Consider the diagram below. Show that the area of $\triangle ABC$ is $\frac{1}{2}ab \sin \theta$.

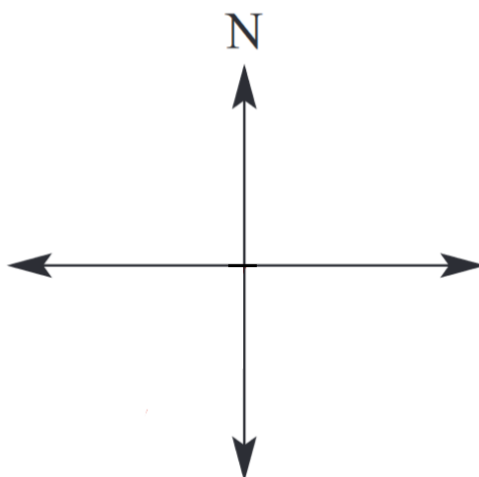
2



4. A helicopter flies on a bearing of 137° for 205 km and then flies due east for 181 km.

(a) Represent this information on the diagram below.

1



(b) How far east (to the nearest km) has the helicopter travelled from its starting point?

2

Short Answer Questions – Show all working.**Probability**

5. Lucy rolls n dice.

(a) Write an expression for the number of possible outcomes. **1**

(b) What is the probability that Lucy rolls *no* sixes? **1**

(c) What is the probability that Lucy rolls *at least* one six? **1**

6. In a certain town there are 10 000 cars, each having a licence plate number from 1 to 10 000. What is the probability that the first car you see on a given day does not have a 6 or 7 in its number plate? **1**

End of examination 😊